

BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

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Abstract

Recovering prior client-specific advice from a legal consultation archive is difficult because dense retrieval may rank a relevant question turn highly while omitting the lawyer’s answer—an *episode incompleteness* failure under same-domain interference. BIMS-LEGAL is a dual-store system for Chinese legal consultation logs: a fast episodic turn index with session identifiers and a slower BIRCH semantic store. Soft O2 performs turn-level session binding by giving siblings attenuated inherited scores (β s) rather than hard score copy. We evaluate exact replay as diagnostic and use paraphrase, follow-up, and advice-recall as primary non-exact channels. BM25-joint is a strong lexical baseline on exact and high-overlap queries, but Soft O2 is more effective on non-exact channels where online turn-level retrieval can recover a partial episode and then bind its answer sibling. Soft O2-C and gated Hybrid are reported as exploratory, gold-assisted solvability bounds for cross-session recovery, not as claims about natural unsupervised deployment. We report corrected graded nDCG alongside Answer Hit and Episode Completeness, with paired significance tests and a failure taxonomy separating complete, incomplete, and missed-session retrieval.

Keywords: legal information retrieval, consultation memory, soft episode binding, dual-store memory, cluster binding, Chinese legal corpora

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1. Introduction

A junior associate may ask a firm assistant: “Last month you advised this client on terminating a fixed-term contract—what remedy did we recommend?” The system returns several near-identical “contract termination” questions, ranks the stored user utterance highly, and never surfaces the partner’s written advice. The subsequent reply may sound fluent yet be grounded in the wrong episode—or in none. In consultation archives, questions and answers share specialized terminology; neighbouring matters are topically adjacent yet legally distinct; and the professionally useful evidence unit is the full consultation *episode*.

Legal informatics has long studied statutory interpretation, case retrieval, and advisory systems Martinez-Gil (2023); Ashley (2017). Recent legal large language models (LLMs) improve statutory and fact-pattern answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical legal information retrieval (IR) mainly targets *external* authority—statutes, cases, or community answers—as documents Askari et al. (2024); Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented dialogue systems recover evidence from evolving conversation logs Wu et al. (2024); Maharana et al. (2024); Packer et al. (2023). The gap addressed here is *internal consultation memory*: retrieving prior client-specific advice from a growing same-domain archive under lexical interference.

BIMS-LEGAL is a dual-store Brain-Inspired Memory System for legal consultation archives, designed and evaluated in this study. The *systemic contribution* is a dual-store architecture that separates a fast episodic pathway (timestamped turn embeddings with speaker role and session identifier *sid*) from a slow semantic pathway based on Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) with deferred consolidation Zhang et al. (1996). Complementary Learning Systems (CLS) theory McClelland et al. (1995); Kumaran et al. (2016) supplies an architectural analogy for this dual-pathway design, not an empirical claim about biological memory. Under same-domain interference, dense turn retrieval in a vector database often returns a question turn while its answer sibling remains outside top- k —a system-level episodic memory failure we term *episode incompleteness*, distinct from a full session miss and exacerbated by IVFPQ score collapse and context fragmentation at moderate archive sizes.

36 The *algorithmic contributions* are soft binding operators that complete
 37 episodes without hard score copy:

- 38 • **Soft O2 (session soft binding).** When a turn is retrieved with score
 39 s , siblings sharing *sid* inherit $\max(s_{\nu}, \beta \cdot s)$, surfacing answer turns
 40 under question-hit / answer-miss while preserving rank competition
 41 among candidates.
- 42 • **Soft O2-C and gated Hybrid (cluster soft binding).** The same
 43 soft-inheritance rule applies within BIRCH clusters when gold evidence
 44 is not co-session with the query; Hybrid triggers cluster binding only
 45 on direct dense hits to avoid displacing same-session siblings.

46 Two **implementation and scale design operators** support rigorous large-
 47 scale shared-store experiments but are not standalone algorithmic contribu-
 48 tions:

- 49 • **O1 (exact FlatIP indexing).** At hundreds-to-thousands of turns,
 50 IVFPQ product quantization is under-trained and collapses Answer
 51 Hit; FlatIP restores faithful turn scores so Soft O2 inherits uncorrupted
 52 sibling signals.
- 53 • **O3 (bulk-load consolidation).** Deferred BIRCH and disk writes
 54 make archive construction tractable without changing retrieval seman-
 55 tics.

56 We ask four research questions:

- 57 • **RQ1.** Under same-domain interference, does episode incompleteness
 58 (question hit / answer miss) dominate full session miss?
- 59 • **RQ2.** Do Soft O2 and Soft O2-C address episode incompleteness under
 60 measured index distortion, and what incremental gains does Soft O2
 61 provide over FlatIP and hard parent hydration?
- 62 • **RQ3.** On the *same* LegalEp /LegalMem stores, when is the slow
 63 (BIRCH) pathway effective? Does Soft O2-C help under standard same-
 64 session gold, and under a fair Mix of same-/cross-session gold?
- 65 • **RQ4.** Across CAIL multi-turn and LegalEp paraphrase /advice-recall
 66 channels, how large and session-tied are Soft O2 gains, and how do
 67 Soft O2 and Soft O2-C /Hybrid complement each other?

Contributions..

1. A dual-store memory architecture for Chinese legal consultation archives, separating fast episodic turn recovery from slower BIRCH semantic integration under same-domain interference.
2. Soft O2, a turn-level session soft-binding operator that addresses episode incompleteness without hard parent hydration or session-max score copy.
3. A conservative three-candidate diagnostic of Soft O2 score competition using the gap ratio $\rho = s_d/s_h$; default $\beta=0.98$ is chosen by validation sweep, not by a closed-form optimum.
4. An honest comparison with strong lexical baselines: BM25-joint and dense joint-episode indexing are highly competitive on exact/surface-overlap channels, while Soft O2 targets paraphrase, follow-up, and advice-recall under an online turn-level index.
5. Exploratory Soft O2-C / gated Hybrid results under a gold-assisted Mix solvability bound, plus corrected nDCG, failure taxonomy, and paired significance reporting.
6. O1 exact FlatIP and O3 bulk-load consolidation as implementation operators enabling the shared-store ablations reported here.

Section 2 reviews related work. Section 3 presents BIMS-LEGAL operators and the Soft O2 diagnostic model. Section 4 details corpora, corrected metrics, and protocols. Section 5 reports results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.1. Legal IR and long-horizon dialogue memory

Legal QA and retrieval systems typically ground generation in statutes, cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware similar-case ranking further show how interaction context and structured legal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024). Those pipelines optimize access to external authority. Consultation memory retrieval targets a different evidence source: *prior consultation episodes* inside an agent’s own archive. Confusing two clients’ histories can produce fluent yet professionally inappropriate responses even when statute retrieval is correct. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM

103 improve statutory and advisory generation Yue et al. (2023); Huang et al.
 104 (2023); Yue et al. (2024), but they still need a reliable memory index when
 105 prior advice must be recovered from a growing same-domain log.

106 Legal consultation archives add a further difficulty: high lexical overlap
 107 among topically adjacent matters, with the professionally useful evidence
 108 unit being the full question-answer episode. This setting motivates *dual-*
 109 *store* memory architectures that keep fast episodic turn retrieval separate
 110 from slower semantic clustering, and session-aware soft binding operators
 111 that recover answer turns under same-domain interference without collapsing
 112 rank competition among unrelated candidates.

113 2.2. Dual-store memory and episodic retrieval

114 Dual-store designs distinguish fast episodic encoding from slower seman-
 115 tic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving
 116 et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dia-
 117 logue agents extend context via hierarchical paging, summary banks, graphs,
 118 and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Ras-
 119 mussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang
 120 et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved
 121 units as flat passages rather than *multi-turn consultation episodes* whose
 122 answer turn may rank below an unrelated question turn sharing legal vo-
 123 cabulary. LongMemEval and LoCoMo evaluate long-horizon recall and QA
 124 over personal conversation logs Wu et al. (2024); Maharana et al. (2024),
 125 while broader memory-agent surveys emphasize persistence beyond the con-
 126 text window Wu et al. (2025); Tan et al. (2025). Recent conversational
 127 RAG work stresses reuse of historical search evidence across turns Mo et al.
 128 (2026). BIMS-LEGAL positions its dual-store contribution at this intersec-
 129 tion: episodic turn embeddings with *sid* for same-session recovery (Soft O2),
 130 and BIRCH centroids with Soft O2-C for cross-session recovery when query
 131 and gold evidence occupy different sessions but share thematic structure
 132 Zhang et al. (1996). Soft O2 directly addresses the episodic retrieval chal-
 133 lenge that parent-document hydration and session-max aggregation handle
 134 via hard score copy: under dense same-domain interference, a question turn
 135 can dominate top- k while its answer sibling remains absent, producing fluent
 136 yet ungrounded downstream generation unless siblings enter under attenu-
 137 ated inherited scores.

138 2.3. Session-aware retrieval

139 Parent-document hydration, session aggregation, and joint session index-
 140 ing are standard session-aware mechanisms for attaching sibling context to a
 141 dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and keeps
 142 ranking competition among candidates. Parent-document and hierarchical
 143 retrieval similarly attach child evidence to a parent unit Callan et al. (1994);
 144 Dai et al. (2019); Zaheer et al. (2017); Karpukhin et al. (2020); Soft O2
 145 differs by using attenuated turn-level score propagation rather than hard hy-
 146 dration or atomic session documents. Lexical BM25 Robertson and Zaragoza
 147 (2009), dense passage retrieval Karpukhin et al. (2020), dense-sparse Recipro-
 148 cal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira
 149 et al. (2019); Chen et al. (2024) provide complementary controls outside ses-
 150 sion membership. Exact flat indexes and product-quantized ANN backends
 151 Johnson et al. (2019) further affect whether sibling binding is applied over
 152 faithful turn scores. Table 1 summarizes the session-structure contrasts used
 153 in the main grids.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	Unit	Expansion	Type	Rank
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+ <i>sid</i>	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft $\beta_c s$	Soft	Yes

154 *On the reported corpora, session-max rankings coincide with hard parent hydration
 155 under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2
 156 denotes session soft binding; Soft O2-C denotes cluster soft binding on the BIRCH path-
 157 way.

158 We reuse CLS vocabulary as an architectural analogy for BIMS-LEGAL:
 159 turn embeddings with timestamps, roles, and session identifiers as episodic
 160 elements; Soft O2 as episode binding on the episodic pathway; BIRCH cen-
 161 troids with Soft O2-C as semantic integration on the slow pathway; and
 162 shared-store distractors as interference. The dual-store split is the systemic
 163 design; Soft O2 and Soft O2-C are the algorithmic mechanisms evaluated on
 164 Chinese legal consultation archives.

165 3. Method: BIMS-LEGAL

166 3.1. Architecture overview

167 Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complemen-
168 tary episodic and semantic stores over the same consultation archive.

169 **Definition 1** (Episodic and semantic elements). *An episodic element is*
170 *$e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, \text{sid}_i)$, where $\mathbf{v}_i \in \mathbb{R}^d$ is an embedding, t_i a timestamp, c_i*
171 *conversational context (including speaker role), ϕ_i surface features, and sid_i*
172 *the consultation session identifier. A semantic element is $s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$*
173 *with centroid $\mathbf{c}_j$, member set C_j , label κ_j , and formation statistics τ_j . Write*
174 *$\text{Sib}(t) = \{t' : \text{sid}_{t'} = \text{sid}_t\} \setminus \{t\}$ for session siblings and $\text{Clus}(t) = \{t' : \text{cid}_{t'} =$*
175 *$\text{cid}_t\} \setminus \{t\}$ for BIRCH-cluster siblings.*

176 **Definition 2** (Dual-store retrieval map). *Given query q with embedding*
177 *$\mathbf{v}_q$, base episodic retrieval returns a scored candidate set $\mathcal{R}_0(q) = \{(t, s_t)\}$*
178 *under Eq. (1). Soft O2 produces $\mathcal{R}_{\text{O2}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Sib}, \beta)$ (Sec-*
179 *tion 3.3). Soft O2-C produces $\mathcal{R}_{\text{O2C}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Clus}, \beta_c)$ (Sec-*
180 *tion 3.5). Gated Hybrid composes session binding then cross-session cluster*
181 *binding (Algorithm 2). The returned list is the top- k of the bound set under*
182 *the rewritten scores.*

183 3.2. Base scoring and implementation operators (O1, O3)

184 For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory
185 turn m by

$$\text{score}(m) = \alpha S(\mathbf{v}_q, \mathbf{v}_m) + (1 - \alpha) R(t_m), \quad (1)$$

186 where S is cosine similarity on ℓ_2 -normalized embeddings and

$$R(t_m) = \max\left(0, 1 - \frac{\Delta_t}{30 \text{ days}}\right), \quad (2)$$

187 with Δ_t the elapsed time between query time and the turn timestamp. Unless
188 stated otherwise, $\alpha=0.7$. Eq. (1) is the ranking score used throughout the
189 legal grids; Soft O2 /Soft O2-C rewrite sibling scores after this fusion step
190 and do not replace Eq. (1).

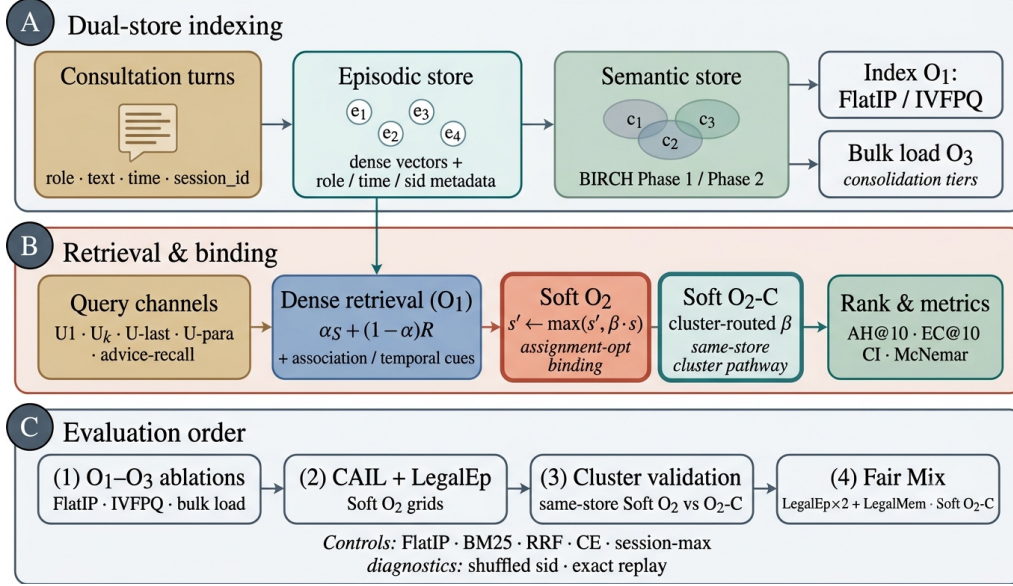


Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and BIRCH semantic clustering (O3 bulk-load consolidation). (B) Query-time dense retrieval under O1 FlatIP with Soft O2 session binding and Soft O2-C cluster binding (algorithmic core); O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, CAIL /LegalEp Soft O2 grids, same-store cluster validation, and fair Mix Soft O2-C.

191 *O1: Exact FlatIP indexing..* Let $\hat{S}$ denote the similarity induced by an ANN
 192 backend. At hundreds to a few thousand turns, FAISS IVFPQ is often under-
 193 trained, so $\|\hat{S} - S\|$ is large enough to reorder turns and break Soft O2
 194 inheritance. O1 forces $\hat{S} = S$ via exact FlatIP (inner-product search on
 195 ℓ_2 -normalized embeddings). Quantization error depresses Answer Hit (Ta-
 196 ble A.8: IVFPQ AH 0.330 at $M=1600$ on LegalEp-DISC exact). O1 is an
 197 engineering prerequisite for rigorous Soft O2 evaluation, not a standalone
 198 retrieval algorithm.

199 *O3: Bulk-load consolidation..* During bulk construction, O3 replaces the per-
 200 insert update ($e_i \mapsto \text{BIRCH}(e_i); \text{Flush}$) by a deferred operator **Finalize** =
 201 $\text{BIRCH}(\{e_i\}_{i=1}^N) \circ \text{Flush}$, executed once after all episodic writes. O3 does
 202 not change retrieval semantics; it makes $M \geq 400$ shared-store experiments
 203 runnable (≈ 11 min wall-clock for $M=400$; amortized ≈ 0.825 s/turn over
 204 ~ 800 turns). Quality-latency curves to $M=6400$ appear in Section 5.6.

Algorithm 1 Soft O2 session soft binding

Require: Query q , base hits $\mathcal{R}_0$, top- k , attenuation β

```
1:  $\mathcal{R} \leftarrow \mathcal{R}_0$ 
2: for each hit  $t \in \mathcal{R}_0$  with score  $s_t$  do
3:   for each sibling  $t' \in \text{Sib}(t)$  absent from  $\mathcal{R}$  do
4:     insert  $(t', \beta \cdot s_t)$  into  $\mathcal{R}$ 
5:   end for
6: end for
7:                                      $\triangleright$  Completeness pass after fusion with Eq. (1)
8: for each session  $sid$  represented in  $\mathcal{R}$  do
9:    $s_{\max} \leftarrow \max\{s_u : u \in \mathcal{R}, sid_u = sid\}$ 
10:  for each  $t' \in \text{Sib}$  of  $sid$  still absent do
11:    insert  $(t', \beta \cdot s_{\max})$  into  $\mathcal{R}$ 
12:  end for
13: end for
14: return top- $k$  of  $\mathcal{R}$  by descending score
```

205 *3.3. Soft O2: session soft binding*

206 *Motivation..* Under same-domain interference, a retrieved question turn can
207 score highly while its answer sibling remains outside top- k —episode incom-
208 pleteness compounded by IVFPQ score collapse and context fragmentation.
209 Soft O2 completes episodes without hard score copy.

210 **Definition 3** (Soft binding operator). *Given a scored set $\mathcal{R} = \{(t, s_t)\}$, a*
211 *sibling relation $\mathcal{N}(\cdot)$, and attenuation $\beta \in (0, 1]$, define*

$$\text{SoftBind}(\mathcal{R}; \mathcal{N}, \beta) : \quad s_{t'} \leftarrow \max(s_{t'}, \beta \cdot s_t) \quad \text{for all } t \in \mathcal{R}, t' \in \mathcal{N}(t). \quad (3)$$

212 *Soft O2 is $\text{SoftBind}(\cdot; \text{Sib}, \beta)$ with default $\beta=0.98$. Hard parent hydration is*
213 *the special case $\beta=1$ with forced insertion of all siblings; session-max replaces*
214 *each member score by $\max_{t \in sid} s_t$.*

215 Algorithm 1 states the deployed Soft O2 procedure (expand + complete-
216 ness truncate), matching the public implementation.

217 *3.4. Soft O2 attenuation as diagnostic score competition*

218 The attenuation β controls how strongly a retrieved turn promotes its
219 session siblings. We treat the following as a *diagnostic* model, not a proof

of a universal optimum. In particular, a zero-margin hinge objective with $\beta \in (0, 1]$ does not identify a meaningful quantile solution: when the rank margin is zero, the rank term is inactive for all $\beta < 1$, so its weight cannot determine a closed-form optimum.

Definition 4 (Three-candidate diagnostic model). *For a query whose dense retrieval directly hits a gold turn h with fused score $s_h > 0$, let a be the gold answer sibling and let d be the strongest non-session distractor in a wide candidate pool, with scores s_a and s_d . Soft O2 rewrites*

$$s'_h = s_h, \quad s'_a = \max(s_a, \beta s_h), \quad s'_d = s_d, \quad (4)$$

and defines the gap ratio $\rho = s_d/s_h$. Because d is the strongest measured distractor, conditions based on ρ are conservative sufficient conditions for beating that distractor, not exact top- k boundary conditions for every competitor.

The binding analysis is most relevant in the soft-injection regime $s_a \leq \beta s_h$, where $s'_a = \beta s_h$.

Proposition 1 (Feasible attenuation band). *Under Definition 4, assume the soft-injection case $s_a \leq \beta s_h$. Then (i) availability against the measured distractor requires $\beta > \rho$; (ii) rank preservation of the direct hit over the injected sibling requires $\beta < 1$, and more generally $s'_a < s_h$; (iii) when $\rho < 1$, any $\beta \in (\rho, 1)$ satisfies both conditions in the active injection case. If $s_a > s_h$ already, rank preservation depends on the observed sibling score rather than on β alone.*

Proof. If $s_a \leq \beta s_h$, then $s'_a = \beta s_h$. Availability follows from $\beta s_h > s_d$, i.e. $\beta > \rho$. Rank preservation requires $s'_a < s_h$, which reduces to $\beta < 1$ when $s_h > 0$. $\square$

Exploratory soft-rank diagnostic. We also minimize an exploratory soft pairwise diagnostic on a validation grid,

$$\mathcal{L}_{\text{sr}}(\beta; \rho) = -\log \sigma(\tau(\beta - \rho)) - \gamma \log \sigma(\tau(1 - \beta)), \quad (5)$$

with logistic σ , temperature τ , and rank weight γ . This visualization is useful for studying the availability–ranking trade-off, but it depends on the diagnostic pool, query channel, and chosen hyperparameters. Default $\beta=0.98$

Table 2: Diagnostic gap-ratio statistics on LegalEp $M=3000$ stores (exact-channel query proxy; strongest measured non-session distractor). These are calibration diagnostics, not derivations of an optimal β .

Corpus	ρ_{50}	ρ_{95}	Cover@0.90	Default β
LegalEp-DISC	0.746	0.854	0.981	0.98
LegalEp-Lawyer	0.662	0.815	0.998	0.98

is selected from the sweep in Table A.6, where $\{0.90, 0.95, 0.98\}$ form a stable high-AH band.

Fair controls include hard parent hydration, session-max aggregation, and shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max coincide under full-sibling expansion; we report a single hard-expansion baseline. Dense joint episode indexing is an atomic-index bound with a different evidence unit.

3.5. Semantic consolidation and Soft O2-C

Semantic consolidation follows a BIRCH-style dual-phase procedure Zhang et al. (1996).

Definition 5 (BIRCH dual-phase consolidation). *Let $\{\mathbf{c}_j\}$ be current centroids and θ_H, θ_M the assign /merge thresholds. **Phase 1 (assign)**. For a new embedding $\mathbf{v}$, set $j^* = \arg \max_j S(\mathbf{v}, \mathbf{c}_j)$; if $S(\mathbf{v}, \mathbf{c}_{j^*}) \geq \theta_H$ then $C_{j^*} \leftarrow C_{j^*} \cup \{\mathbf{v}\}$ and refresh $\mathbf{c}_{j^*}$, else spawn a new cluster. **Phase 2 (merge)**. Periodically, merge pairs with $S(\mathbf{c}_i, \mathbf{c}_j) \geq \theta_M$ and purge near-duplicates above a redundancy threshold. Under O3, both phases run inside **Finalize**. Each turn t receives a cluster id cid_t , inducing $\text{Clus}(t)$ in Definition 1.*

Definition 6 (Soft O2-C). *Soft O2-C is $\text{SoftBind}(\cdot; \text{Clus}, \beta_c)$ with default $\beta_c=0.95 < \beta$, reflecting noisier thematic neighbourhoods than session membership. A per-cluster sibling budget B_c caps $|\text{Clus}(t) \cap \text{Injected}|$ so large thematic clusters cannot flood top- k .*

Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings relative to sessions already covered by dense /Soft O2 hits, and triggers cluster binding only on direct dense hits. This prevents thematic confusers from displacing same-session siblings already recovered by Soft O2. Soft O2 and Soft O2-C are complementary: Soft O2 is the default binder when query

Algorithm 2 Gated Hybrid Soft O2 + Soft O2-C

Require: Base hits $\mathcal{R}_0$, budgets (k, B_c) , attenuations (β, β_c)

```
1:  $\mathcal{R} \leftarrow \text{SoftBind}(\mathcal{R}_0; \text{Sib}, \beta)$  ▷ session pathway first
2:  $\mathcal{S}_{\text{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$ 
3:  $\mathcal{T}_{\text{trig}} \leftarrow \{t \in \mathcal{R}_0\}$  ▷ direct dense hits only
4: for each trigger  $t \in \mathcal{T}_{\text{trig}}$  do
5:    $s_{\text{max}} \leftarrow \max\{s_u : u \in \mathcal{R}, cid_u = cid_t\}$ 
6:   for each  $t' \in \text{Clus}(t)$  with  $sid_{t'} \notin \mathcal{S}_{\text{cov}}$ , up to budget  $B_c$  do
7:      $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\text{max}})$ 
8:   end for
9: end for
10: return top- $k$  of  $\mathcal{R}$ 
```

276 and gold share sid ; Soft O2-C is required when gold evidence is held in a
277 different session but the same cluster.

278 4. Corpora and Evaluation Protocol

279 Figure 2 sketches the shared-corpus design used throughout the legal eval-
280 uation. Gold needles and same-domain distractors occupy one store. Each
281 query probes that shared archive under fixed interference conditions. Pri-
282 vately rebuilt haystacks are avoided so that system comparisons isolate the
283 retrieval operator.

284 4.1. Datasets and query channels

285 Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues.
286 Gold needles are authentic preliminary and final dialogues, with conversation
287 turns completed by label turns where applicable. Query channels cover the
288 first user turn (U1), a later in-dialogue user turn (Uk-followup), the last user
289 turn (U-last), and constrained paraphrases (U-pa). Evidence defaults to
290 assistant turns in the target session.

291 LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and
292 Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is
293 a natural (question, answer) pair with session identifier sid . Reconstruction
294 applies length bounds and legal-content filters, excludes exam prompts, re-
295 moves near-duplicates, separates needles from distractors under a fixed seed,
296 and retains an audit subsample. After filtering, each LegalEp corpus retains

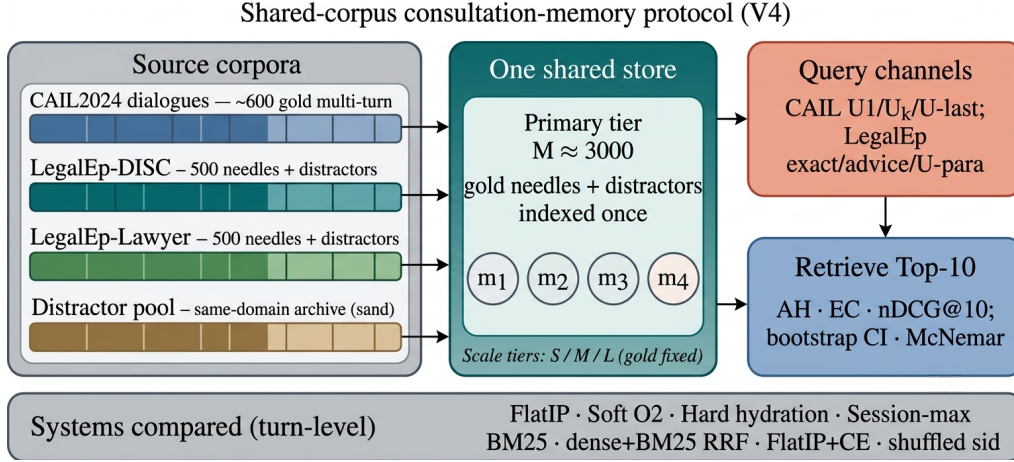


Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

297 3500 passed sessions (500 needles, 2500 distractors at the $M=3000$ primary
298 tier).

299 *Query-channel policy (addressing exact-match shortcuts)*.. Exact replay uses
300 the stored user question and is reported only as a *diagnostic* upper bound
301 on surface-form recovery. Primary claims use paraphrase, U_k -followup / $U\text{-}$
302 last, and advice-recall queries whose surface form is not identical to the
303 stored question text; paraphrase and advice-recall templates are generated
304 under a fixed seed and audited on a subsample. LegalMem-MT reuses CAIL
305 multi-turn gold under the same store for same-session Soft O2 vs Soft O2-C
306 contrasts and for Mix reconstructions.

307 Archive size is varied with gold needles held fixed. Unless stated otherwise,
308 Soft O2 transfer tables use the $M \approx 3000$ tier; O1–O2 ablations use $M=400$,
309 $S=300$.

310 4.2. Metrics and baselines

311 Primary metrics are Answer Hit@ k (AH), Episode Completeness@ k (EC),
312 and corrected normalized Discounted Cumulative Gain@ k (nDCG) at $k=10$.
313 AH is binary: whether a gold answer turn appears in the top- k . EC is the
314 mean coverage of gold turns within the target session. Session Hit@ k is
315 reported only for diagnosis. For nDCG we use fixed graded relevance over

the full gold session: answer turns $rel=1.0$, other same-session turns $rel=0.5$, all other turns $rel=0.0$. The ideal denominator (IDCG) is computed from all gold-relevant turns in the target session and truncated at k , so every system on a query shares the same IDCG. Earlier draft values based on retrieved-item IDCG are not comparable and have been recomputed (artifact `paper/ipm/figures/corrected_metrics_*.json`). Primary contrasts use bootstrap 95% CIs and McNemar mid- p tests on paired per-query hits; where families of related hypotheses are interpreted jointly, we also report Holm-corrected p -values (Table A.7).

Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies soft sibling score inheritance on the episodic pathway. Soft O2-C applies the analogous rule within BIRCH clusters. Hard parent hydration expands a hit to all siblings by copying the trigger score. Lexical controls include BM25 over turns and over joint question-answer documents, together with dense+BM25 Reciprocal Rank Fusion (RRF). A cross-encoder (CE) control reranks the top-30 FlatIP candidates with `BAAI/bge-reranker-v2-m3`. Negative and sensitivity controls include shuffled *sid*, a β sweep, and IVFPQ as a quantization reference.

4.3. Fair Mix protocol for Soft O2-C

On fully same-session LegalMem grids, Soft O2 already recovers gold answers that share *sid* with the query, so Soft O2-C has little exclusive headroom and may inject thematic confusers. We therefore reconstruct the *same* LegalEp and LegalMem archives under a Mix protocol: 70% of gold episodes are split into query-side S_a and evidence-side S_b with distinct session identifiers, while 30% remain intact. Distractors are unchanged. All operators—FlatIP, Soft O2, Soft O2-C, and gated Hybrid—share unrestricted dense retrieval on this store; we report overall AH and stratum AH on `cross_session` vs `same_session` queries.

Glue as a solvability bound (not a retrieval cheat).. After BIRCH consolidation on the Mix store, each cross-session S_a/S_b split pair is *glued* into one cluster identifier. This step uses gold split metadata available only in the evaluation protocol—it does *not* inject labels at query time and is *not* used in real unsupervised clustering deployments. Its sole purpose is to establish a **solvability bound**: if Soft O2-C cannot recover cross-session gold when the semantic store *must* contain both halves of a split pair in one cluster,

then failure reflects mechanism limits rather than accidental cluster separation. Glue therefore tests whether cluster soft binding *can* complete episodes when the slow pathway has the requisite thematic co-membership; natural same-cluster rates without glue are lower and are reported only as diagnostics (Section 6.3). Soft O2’s session membership constraint is unchanged—glue affects only whether Soft O2-C has a cluster in which to bind, not whether Soft O2 receives privileged label access. LegalEp Mix uses advice-recall queries; LegalMem Mix uses mid-split Uk-followup queries. External LongMemEval /LoCoMo numbers from prior retrieval runs on this codebase (QA correctness 70.7% / 68.2% on released test splits) are cited only as long-horizon memory context Wu et al. (2024); Maharana et al. (2024), not as Soft O2-C evidence on legal corpora.

5. Experiments

Unless noted, the embedding model is `nomic-embed-text-v2-moe` Nussbaum et al. (2025), $k=10$, seed 42. We report primary contrasts in figures and keep full numeric grids in the appendix. Result files and scripts are released with the accompanying repository so that each table maps to a public JSON artifact.

5.1. Implementation operators and Soft O2 ablation

Table 3 reports the shared-store ablation on DISC-Law and Lawyer-LLaMA ($M=400$, $S=300$). IVFPQ baseline Answer Hit is 0.540 / 0.397. O1 alone yields a small gain on DISC (0.567) and a near-tie on Lawyer (0.397), confirming that exact indexing is a necessary implementation prerequisite but insufficient alone. O1+Soft O2 raises Answer Hit to 0.780 / 0.680 (+0.240 / +0.283 over baseline) and Episode Completeness to 0.888 / 0.840—the primary algorithmic gain. O3 is enabled for all configurations in this ladder as a construction prerequisite; it does not alter retrieval scores.

5.2. RQ1 failure taxonomy

We classify each query into *complete* (answer evidence in top- k), *incomplete* (target session partially retrieved but answer missing), or *session miss* (no target-session turn in top- k). Table 4 shows that incomplete retrieval is a substantial recurrent failure mode for FlatIP on primary non-exact channels, motivating Soft O2.

Table 3: O1+Soft O2 ablation on the shared store ($M=400$, $S=300$, $k=10$). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus.

Corpus	System	EC@10	AH@10	nDCG@10
DISC-Law	IVFPQ baseline	0.770	0.540	0.782
	+ O1 FlatIP	0.782	0.567	0.790
	+ O1+O2 Soft O2	0.888	0.780	0.889
Lawyer-LLaMA	IVFPQ baseline	0.698	0.397	0.733
	+ O1 FlatIP	0.698	0.397	0.733
	+ O1+O2 Soft O2	0.840	0.680	0.856

Table 4: RQ1 failure taxonomy (FlatIP, $k=10$). Complete $\approx$ Answer Hit; incomplete $\approx$ Session Hit – Answer Hit (answer-only cases are rare); session miss $\approx 1 - \text{Session Hit}$.

Corpus / channel	System	Complete	Incomplete	Session miss
LegalEp-DISC / u-para	FlatIP	54.7%	36.0%	9.3%
LegalEp-DISC / advice	FlatIP	34.2%	26.6%	39.2%
LegalEp-Lawyer / u-para	FlatIP	56.8%	39.6%	3.6%
CAIL / Uk-followup	FlatIP	27.0%	69.8%	3.2%

384 5.3. Strong lexical and joint-episode baselines

385 BM25-joint indexes each question–answer episode as one lexical document
386 and is therefore very strong on exact replay and other high-overlap channels.
387 Soft O2 instead assumes an online turn-level index and uses session bind-
388 ing to recover missing answer turns after a dense hit. Table 5 reports this
389 trade-off on the same store, query set, and $k=10$. Soft O2 is not claimed
390 to dominate BM25-joint universally; its advantage is on non-exact channels
391 and on preserving turn-level ranking flexibility.

392 5.4. Soft O2 on session (CAIL and LegalEp)

393 Figure 3 summarizes AH@10 and EC@10 on authentic CAIL multi-turn
394 channels at $M \approx 3000$ under O1 FlatIP. Soft O2 lifts Answer Hit over FlatIP
395 on every channel (U1 0.788 vs 0.570*; Uk 0.698 vs 0.270*; U-last 0.762 vs
396 0.245*). Shuffled *sid* removes the gain (U1 0.452; Uk 0.230; U-last 0.200),
397 tying the improvement to correct episode membership (RQ1–RQ2). Hard
398 expansion can raise AH slightly above Soft O2 (e.g., U1 0.802 vs 0.788) while
399 lowering EC (U1 0.578 vs Soft O2 0.664). Soft O2 therefore behaves as a

Table 5: BM25-joint / joint-QA vs Soft O2 (AH@10, primary tier). Joint baselines are strongest on exact/surface overlap; Soft O2 targets paraphrase and advice-recall.

Corpus / channel	BM25-joint	Joint QA	Soft O2
CAIL / U1	0.993	—	0.788
CAIL / Uk	0.840	—	0.698
CAIL / U-last	0.823	—	0.762
LegalEp-DISC / exact	0.976	—	0.638
LegalEp-DISC / u-para	0.592	—	0.648
LegalEp-DISC / advice	0.432	—	0.428
LegalEp-Lawyer / exact	—	—	0.752
LegalEp-Lawyer / u-para	—	—	0.759
LegalEp-Lawyer / advice	0.652	—	0.472

CAIL2024 multi-turn results ($M \approx 3000$, turn-level systems)

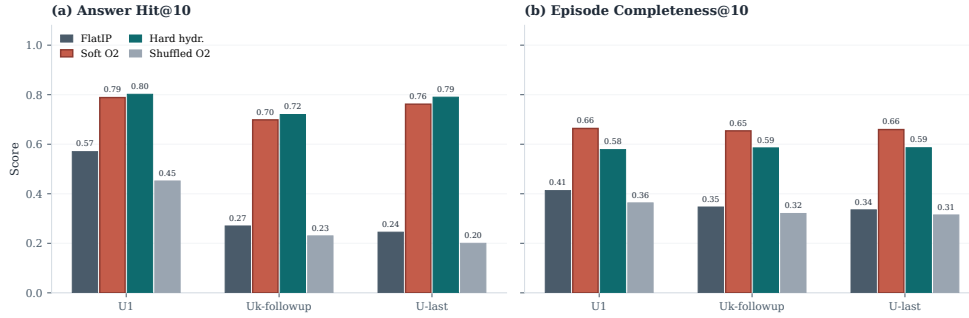


Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard expansion can maximize AH at a modest EC cost.

rank-preserving binder: siblings enter under attenuated scores, and episode
coverage remains higher than under hard score copy. On CAIL, nDCG often
falls relative to FlatIP even as AH and EC rise, reflecting an availability-
ranking trade-off. Soft O2 still exceeds FlatIP+CE on every CAIL channel
(Table A.5). Full rows appear in Table A.1.

Figure 4 reports LegalEp transfer. On LegalEp-DISC exact replay Soft O2
and FlatIP stay near-tied (0.638 vs 0.640), where both turns are already
recoverable for many queries—consistent with treating exact as diagnostic.
On Lawyer exact Soft O2 improves AH to 0.752 from FlatIP 0.692 (McNe-
mar mid- $p=0.024^*$). Paraphrase yields the clearest LegalEp gains: Soft O2
reaches 0.648 vs FlatIP 0.547 on DISC and 0.759 vs 0.568 on Lawyer, ex-

LegalEp ($M = 3000$): Soft O2 gains are strongest on paraphrase and remain positive on advice-recall

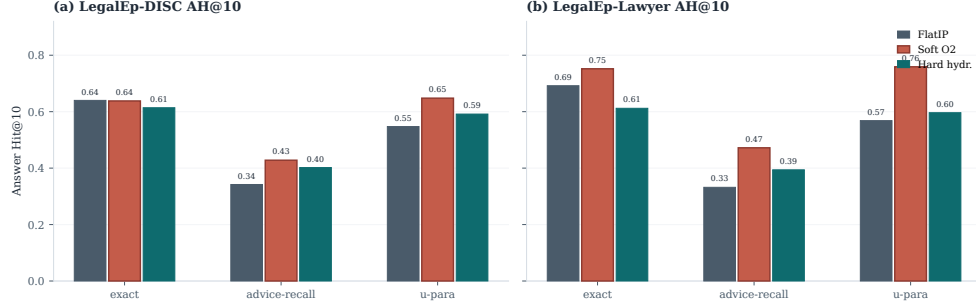


Figure 4: LegalEp AH@10 ($M=3000$, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

ceeding hard expansion on both corpora (mid- $p < 0.001^{**}$ on both corpora). Advice-recall queries omit the stored question surface form: Soft O2 raises AH to 0.428 from FlatIP 0.342 on DISC and to 0.472 from 0.332 on Lawyer, again above hard expansion (mid- $p < 0.001^{**}$), while shuffled *sid* falls toward or below FlatIP. Detailed grids are in Tables A.2–A.3.

5.5. Same-store cluster pathway validation

Table 6 contrasts Soft O2 and ungated Soft O2-C on standard same-session LegalEp /LegalMem stores (unrestricted dense; gold answers share *sid* with the query). Soft O2 dominates Soft O2-C on every reported channel (e.g., LegalMem U1 AH 0.903 vs 0.712; LegalEp-DISC exact 0.836 vs 0.744). The slow pathway is therefore *not* a free upgrade when episode membership already encodes the gold answer: cluster siblings inject thematic confusers that displace session-complete evidence.

Table 7 reports the complementary fair Mix setting (RQ3–RQ4): the same stores with 70% cross-session Sa/Sb gold and 30% intact same-session gold, unrestricted dense for every operator. Gated Hybrid Soft O2+Soft O2-C improves over Soft O2 on LegalEp-Lawyer overall (AH 0.534 vs 0.496; mid- $p = 0.010^{*}$) and on the cross-session stratum (0.523 vs 0.446; mid- $p < 10^{-4^{**}}$). On LegalEp-DISC, Hybrid is directionally above Soft O2 (0.486 vs 0.482) but not significant at $N=500$. On LegalMem-MT, Hybrid attains the highest overall AH (0.646 vs Soft O2 0.618; mid- $p = 0.18$; cross-session mid- $p = 0.095$) while preserving most same-session hits (0.867 vs 0.887). Ungated Soft O2-C can raise cross-session hits but often displaces same-session siblings; Hybrid

Table 6: Same-session Soft O2 vs Soft O2-C on the shared LegalEp /LegalMem stores (AH@10). Soft O2-C is ungated cluster soft binding. Boldface: better AH within each row.

Corpus / channel	Soft O2	Soft O2-C
LegalEp-DISC / exact	0.836	0.744
LegalEp-DISC / u-para	0.807	0.730
LegalEp-DISC / advice	0.548	0.490
LegalEp-Lawyer / exact	0.818	0.710
LegalEp-Lawyer / u-para	0.817	0.733
LegalEp-Lawyer / advice	0.546	0.448
LegalMem-MT / U1	0.903	0.712
LegalMem-MT / Uk	0.793	0.385

triggers cluster binding only on direct dense hits. Together, Tables 6–7 show that the BIRCH pathway is effective precisely when gold evidence is not co-session with the query.

McNemar mid- p tests on paired Answer Hit; * $p < 0.05$, ** $p < 0.01$.

5.6. Controls, significance, scale, and QA audit

Lexical BM25 is strong on exact replay (DISC exact BM25-turn 0.802, BM25-joint 0.976) and weaker on advice-recall (0.382 / 0.432). Dense+BM25 RRF complements Soft O2 on surface-overlap channels. Cross-encoder reranking (`bge-reranker-v2-m3`, top-30) improves FlatIP inside the CE suite, yet remains below Soft O2 on CAIL (Uk FlatIP+CE 0.375 vs Soft O2 0.698; Table A.5). A β sweep peaks near $\{0.9, 0.95, 0.98\}$ and weakens at $\beta=0.5$ (Table A.6), consistent with the diagnostic gap-ratio band in Section 3.4 (Table 2). McNemar mid- p tests (Table A.7) show Soft O2 versus FlatIP highly significant on all CAIL channels and on LegalEp paraphrase /advice-recall.

Figure 5 and Table A.8 report quality and warm-query latency on LegalEp-DISC exact ($Q=100$, three repeats; $M \in \{100, 400, 1600, 6400\}$). Soft O2 retains an AH advantage at each tier; IVFPQ collapses at $M=1600$ (AH 0.330), motivating O1; absolute latency grows with M . We treat O1 FlatIP as an engineering prerequisite for Soft O2 at these scales and avoid claiming deployment readiness beyond the measured curve.

End-to-end QA audit (summary).. A dual-protocol QA audit ($N=270$ per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`) links retrieval quality to downstream answerability under a *single* retrieval configuration

Table 7: Fair Mix protocol (same store, unrestricted dense; AH@10). Gold is a 70%/30% mix of cross-session Sa/Sb splits and intact same-session episodes. Soft O2-C denotes ungated cluster binding; Hybrid is Soft O2+gated Soft O2-C. Boldface: best AH within each corpus. Significance vs Soft O2: * $p<0.05$, ** $p<0.01$.

Corpus	System	AH	AH _{cross}	AH _{same}	mid- p vs Soft O2
LegalEp-DISC Mix	FlatIP	0.478	0.477	0.480	—
	Soft O2	0.482	0.443	0.573	—
	Soft O2-C	0.410	0.443	0.333	0.0004**
	Hybrid	0.486	0.469	0.527	0.8099
LegalEp-Lawyer Mix	FlatIP	0.462	0.454	0.480	—
	Soft O2	0.496	0.446	0.613	—
	Soft O2-C	0.464	0.491	0.400	0.0857
	Hybrid	0.534*	0.523	0.560	0.0105*
LegalMem-MT Mix	FlatIP	0.534	0.526	0.553	—
	Soft O2	0.618	0.503	0.887	—
	Soft O2-C	0.500	0.537	0.413	0.0000**
	Hybrid	0.646	0.551	0.867	0.1837

(Soft O2); we do not claim causal superiority over FlatIP in generation without paired downstream controls. Full protocol and stratified results appear in Section 6.2. On an older revision-protocol store ($M=400$, five seeds), Soft O2 AH on LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040 .

6. Discussion

6.1. Findings and scope

RQ1–RQ2 show that episode incompleteness is a substantial recurrent failure mode under same-domain interference (Table 4). O1 restores score fidelity as an implementation prerequisite; Soft O2 supplies the primary algorithmic gain through soft sibling inheritance; O3 enables archive construction at scale. The diagnostic gap-ratio analysis (Section 3.4) explains why β should remain close to but below unity, and why the validation sweep peaks in $\{0.9, 0.95, 0.98\}$ rather than at hard hydration ($\beta=1$). On CAIL, Soft O2 raises Answer Hit by large margins (U1 +0.218, Uk +0.428, U-last +0.517) with collapsing shuffled-*sid* controls, and keeps higher EC than hard expansion despite occasional AH ties. LegalEp paraphrase and advice-recall—not exact replay—carry the primary Soft O2 transfer claims. RQ3 shows that

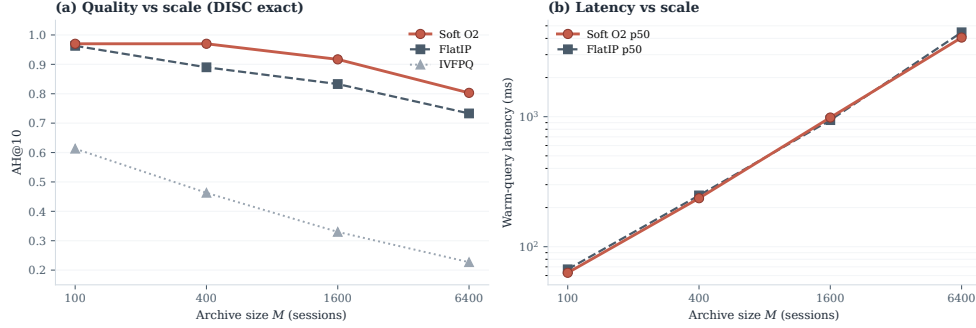


Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M .

474 the BIRCH slow pathway benefits Soft O2-C only when gold is not already
 475 co-session; Mix Hybrid Soft O2+Soft O2-C recovers that regime without
 476 seed-forcing Soft O2 to zero. RQ4: Soft O2 (session) and Soft O2-C /Hy-
 477 brid (cluster) are complementary operators within one BIMS-LEGAL dual-
 478 store. Sparse, fusion, and cross-encoder baselines situate Soft O2 among
 479 standard IR alternatives Robertson and Zaragoza (2009); Cormack et al.
 480 (2009); Nogueira et al. (2019).

481 6.2. RAG application perspective

482 Consultation-memory retrieval is judged downstream by whether an LLM
 483 can answer from recovered evidence rather than confabulate. We audit this
 484 end-to-end with a dual-protocol pipeline on Soft O2 retrieval only: genera-
 485 tor `qwen3:14b` and independent judge `qwen3:32b` are distinct models. Per
 486 corpus we pool $N=270$ judged instances as P1 $n=120$ plus P2 $n=150$ (folder
 487 `legal_qa_n270`). Table 8 summarizes pooled Evidence-Answerability Rate
 488 (EAR) and stratified independent-judge scores. When Soft O2 retrieves com-
 489 plete episodes—both question and answer turns in top- k —the generator re-
 490 ceives grounded prior advice and judged answerability rises in this single-
 491 system audit (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95%
 492 CI $[0.83, 0.90]$ / $[0.82, 0.89]$). Follow-up queries remain the hardest stratum
 493 (independent-judge EAR 0.517 / 0.480), consistent with episodic incomplete-
 494 ness persisting even when surface-form overlap is low.

495 A blind two-annotator legal audit ($n=60$ /corpus) yields Cohen’s
 496 $\kappa=0.71$ / 0.68 and separates retrieval from generation failures (retrieval-
 497 failure share 0.283 / 0.317). Wrong-session contamination rates stay low

498 (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 is
 499 associated with lower rates of a common RAG failure mode in which the
 500 model answers from a neighbouring client’s matter. Hallucinated-legal-basis
 501 rates (0.117 / 0.133) remain non-zero, indicating that grounding in retrieved
 502 episodes does not guarantee statutory correctness—a scope boundary for
 503 deployment. AH and EAR measure evidence availability and judged answer-
 504 ability respectively; statutory correctness, citation validity, and professional
 505 liability lie outside this evaluation.

Table 8: End-to-end QA audit ($N=270$ per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). EAR: Evidence-Answerability Rate (judge score ≥ 0.7). Stratified judge scores: $n=90$ /corpus across exact, paraphrase, and follow-up protocols.

Corpus	Pooled EAR	Stratified judge	Human κ
DISC-Law	0.867	0.757	0.71
Lawyer-LLaMA	0.859	0.729	0.68

506 6.3. Limitations

507 LegalEp episodes are consultation pairs; multi-turn authenticity remains
 508 reserved for CAIL. Mix glue establishes a solvability bound for Soft O2-C
 509 evaluation (Section 4.3); natural same-cluster rates without glue are lower
 510 and are diagnostics only—real unsupervised BIRCH clustering would not
 511 receive split-pair metadata at build time. Soft O2 depends on correct ses-
 512 sion identifiers; Soft O2-C depends on cluster quality. The β diagnostic is
 513 semi-empirical: it assumes a dominant distractor and a well-defined hit score,
 514 and the fitted ρ uses a stored user-turn embedding as an exact-channel query
 515 proxy; paraphrase /advice-recall gap laws may shift the feasible band slightly
 516 but leave the high- β operating region unchanged. LLM paraphrases need hu-
 517 man audit. Statutory correctness, citation validity, and professional liability
 518 remain unevaluated. Multiple channels inflate comparison multiplicity; we
 519 therefore report Holm-corrected p -values for the primary Soft O2 vs FlatIP
 520 family (Table A.7) alongside uncorrected McNemar mid- p tests.

521 *Latency and deployment scope..* Scale curves cover $M \in$
 522 $\{100, 400, 1600, 6400\}$ under a fixed hardware stack (Table A.8). At
 523 $M=6400$, warm-query p50 latency exceeds 4 s (4061 ms for Soft O2; 4458 ms
 524 for FlatIP)—unacceptable for millisecond interactive search. BIMS-LEGAL

should therefore be positioned as **high-precision offline or quasi-real-time consultation memory recovery**—e.g., pre-meeting case review, audit replay, or batch RAG over a firm’s advice archive—not as a sub-millisecond search engine. Future work will combine approximate nearest-neighbour indexes (e.g., HNSW) with Soft O2 and Soft O2-C binding layered on top of faithful candidate retrieval, preserving episode completion while reducing query latency. Behavior at roughly 10^5 sessions is not measured, and we do not claim asymptotic scalability from O3 wall-clock alone.

7. Conclusion

BIMS-LEGAL targets consultation-memory retrieval under same-domain interference in Chinese legal archives through a dual-store design: an episodic turn index paired with a BIRCH semantic store. Soft O2 session soft binding and Soft O2-C with gated Hybrid cluster binding are the algorithmic core, with Soft O2 attenuation β validated on a diagnostic gap-ratio model; O1 exact indexing and O3 bulk-load consolidation are implementation operators that enable the shared-store ablations reported here. Under a shared-corpus protocol spanning CAIL multi-turn dialogues and rebuilt LegalEp episode corpora, Soft O2 improves Answer Hit over FlatIP on multi-turn, paraphrase, and advice-recall channels with collapsing shuffled-*sid* controls, and yields higher episode completeness than hard expansion on CAIL. Same-store Soft O2-C underperforms Soft O2 when gold shares *sid*; under fair Mix reconstructions, gated Hybrid Soft O2+Soft O2-C recovers cross-session gold while Soft O2 remains strong on intact same-session gold. End-to-end QA audit links complete episode retrieval to judged answerability and low wrong-session contamination in downstream generation. Sparse, fusion, and cross-encoder baselines situate Soft O2 among standard IR alternatives.

Declarations

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Computational resources were provided by the Data Law Laboratory, China University of Political Science and Law.

Conflict of interest

The authors declare no competing interests.

558 *Ethics approval*

559 This study uses publicly released research corpora and excludes confiden-
560 tial client data. The system is a research prototype for consultation-memory
561 retrieval.

562 *Data availability*

563 Code, processed evaluation corpora, and primary result files are publicly
564 available at <https://github.com/Kilimajaro/soft-episode-binding-legal-memory>. Upstream legal corpora: CAIL2024 consultation tracks;
565 Hugging Face ShengbinYue/DISC-Law-SFT and Skepsun/lawyer_llama_d
566 ata.

568 *Code availability*

569 The BIMS-LEGAL implementation (O1 FlatIP, Soft O2, O3 bulk-load,
570 Soft O2-C), LegalMem/LegalEp pipelines, Mix builders, and table/figure
571 scripts are released in the same repository with JSON artifacts for each main
572 table.

573 *Author contributions*

574 L.X. (University of Winnipeg) designed BIMS-LEGAL and the
575 Soft O2 /Soft O2-C evaluation, implemented the system, ran the experi-
576 ments, and wrote the manuscript. L.H. (corresponding author; CUPL Data
577 Law Lab and Institute for Data Law) provided methodology guidance, revi-
578 sion, and supervision.

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697 **Appendix A. Numeric grids and controls**

698 Main Soft O2 claims are visualized in Figures 3–5; O1–O2 and Mix
699 Soft O2-C tables appear in the main text. Tables below retain full Soft O2
700 numeric rows for reproducibility.

Table A.1: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid- $p < 0.01$. nDCG@10 uses fixed-gold IDCG (Section 4); regenerate appendix nDCG with the released `recompute_corrected_metrics.py` script when updating artifacts.

Channel	System	AH@10	EC@10	nDCG@10	95% CI (AH)
U1	FlatIP	0.570	0.413	0.805	[0.530,0.610]
	Soft O2	0.788	0.664	0.772	[0.757,0.820]
	Hard hydr.	0.802	0.578	0.840	[0.768,0.835]
	Shuffled O2	0.452	0.363	0.664	[0.410,0.492]
Uk-followup	FlatIP	0.270	0.347	0.884	[0.237,0.307]
	Soft O2	0.698	0.654	0.786	[0.660,0.735]
	Hard hydr.	0.720	0.585	0.841	[0.685,0.753]
	Shuffled O2	0.230	0.320	0.718	[0.198,0.262]
U-last	FlatIP	0.245	0.335	0.891	[0.212,0.280]
	Soft O2	0.762	0.659	0.779	[0.725,0.797]
	Hard hydr.	0.790	0.586	0.844	[0.757,0.822]
	Shuffled O2	0.200	0.315	0.726	[0.172,0.232]

Table A.2: LegalEp-DISC ($M=3000$, 500 needles). Boldface: best AH@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP, McNemar mid- p). Hard hydration coincides with session-max aggregation; only hard hydration is shown.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.640	0.741	—
	Soft O2	0.638	0.780	—
	Hard hydr.	0.614	0.598	—
	Shuffled O2	0.448	0.645	—
advice-recall	FlatIP	0.342	0.438	0.506
	Soft O2	0.428	0.503	0.419
	Hard hydr.	0.402	0.405	0.474
	Shuffled O2	0.310	0.421	0.402
u-param	FlatIP	0.547	0.681	0.705
	Soft O2	0.648	0.735	0.583
	Hard hydr.	0.592	0.571	0.651
	Shuffled O2	0.469	0.610	0.558

Table A.3: LegalEp-Lawyer ($M=3000$, 500 needles). Boldface: best AH@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.692	0.806	0.844
	Soft O2	0.752	0.848	0.687
	Hard hydr.	0.612	0.615	0.754
	Shuffled O2	0.570	0.747	0.653
advice-recall	FlatIP	0.332	0.464	0.589
	Soft O2	0.472	0.540	0.460
	Hard hydr.	0.394	0.403	0.488
	Shuffled O2	0.348	0.467	0.436
u-param	FlatIP	0.568	0.718	0.834
	Soft O2	0.759	0.819	0.645
	Hard hydr.	0.596	0.594	0.707
	Shuffled O2	0.592	0.712	0.627

Table A.4: BM25 turn/joint and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported cells.

Corpus	Channel	BM25-turn	BM25-joint
LegalEp-DISC	exact	0.802	0.976
LegalEp-DISC	advice-recall	0.382	0.432
LegalEp-DISC	exact (dense RRF)	0.780	—
LegalEp-DISC	advice (dense RRF)	0.394	—
CAIL	U1 / Uk / U-last	0.748 / 0.482 / 0.413	0.993 / 0.840 / 0.823
LegalEp-Lawyer	exact / advice	0.784 / 0.460	0.998 / 0.652

Table A.5: Cross-encoder control (AH@10). FlatIP and FlatIP+CE are paired in the CE suite (**bge-reranker-v2-m3** over top-30 dense candidates). Boldface: best AH within each row. Soft O2 reproduces the primary evaluation; FlatIP here may differ slightly from primary FlatIP due to independent CE-suite re-indexing.

Corpus / channel	FlatIP	FlatIP+CE	Soft O2
CAIL / U1	0.585	0.688	0.788
CAIL / Uk	0.265	0.375	0.698
CAIL / U-last	0.228	0.342	0.762
LegalEp-DISC / exact	0.572	0.722	0.638
LegalEp-Lawyer / exact	0.636	0.708	0.752

Table A.6: Soft O2 AH@10 versus β on the primary shared-corpus evaluation. Peak performance lies in $\{0.9, 0.95, 0.98\}$; aggressive attenuation ($\beta=0.5$) weakens binding.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.7: McNemar mid- p on paired Answer Hit (Soft O2 vs FlatIP / hard hydration / FlatIP+CE). Holm-adjusted p is reported for the nine primary Soft O2 vs FlatIP contrasts.

Corpus / channel	Contrast	ΔAH	mid- p	Holm p
CAIL / Uk	O2 vs FlatIP	+0.428	8.99e-48	8.09e-47
CAIL / Uk	O2 vs Hard hydr.	-0.022	0.243	—
CAIL / U1	O2 vs FlatIP	+0.218	5.19e-18	4.67e-17
CAIL / U1	O2 vs Hard hydr.	-0.013	0.435	—
CAIL / U-last	O2 vs FlatIP	+0.517	5.51e-65	4.96e-64
CAIL / U-last	O2 vs Hard hydr.	-0.028	0.0982	—
CAIL / Uk	O2 vs FlatIP+CE	+0.323	2.50e-26	—
CAIL / U1	O2 vs FlatIP+CE	+0.100	3.51e-05	—
CAIL / U-last	O2 vs FlatIP+CE	+0.420	3.92e-41	—
LegalEp-DISC / exact	O2 vs FlatIP	-0.002	0.934	0.934
LegalEp-DISC / exact	O2 vs Hard hydr.	+0.024	0.302	—
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.060	0.0239	0.191
LegalEp-Lawyer / exact	O2 vs Hard hydr.	+0.140	3.50e-10	—
LegalEp-DISC / u-para	O2 vs FlatIP	+0.101	7.95e-05	6.36e-04
LegalEp-DISC / u-para	O2 vs Hard hydr.	+0.056	0.035	—
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.191	2.10e-15	1.68e-14
LegalEp-Lawyer / u-para	O2 vs Hard hydr.	+0.163	7.57e-09	—
LegalEp-DISC / advice	O2 vs FlatIP	+0.086	1.25e-04	8.75e-04
LegalEp-DISC / advice	O2 vs Hard hydr.	+0.026	0.215	—
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.140	1.18e-10	9.44e-10
LegalEp-Lawyer / advice	O2 vs Hard hydr.	+0.078	5.72e-04	—

Table A.8: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	Mode	AH@10	Build (s)	p50 (ms)	p95 (ms)
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	25.2	40	54
100	Soft O2	0.970	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	171	204
400	Soft O2	0.970	30.9	236	313
1600	FlatIP	0.833	359.8	941	1278
1600	IVFPQ	0.330	251.4	669	751
1600	Soft O2	0.917	393.7	985	1172
6400	FlatIP	0.733	4722.8	4458	4969
6400	IVFPQ	0.227	1072.1	2182	3021
6400	Soft O2	0.803	5245.7	4061	4503